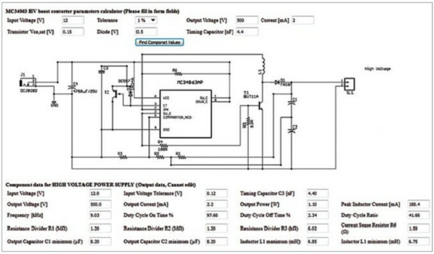

Fig. 6: Screenshot of the program output

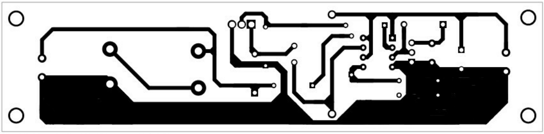

Fig. 7: Actual-size PCB layout of HVDC power supply

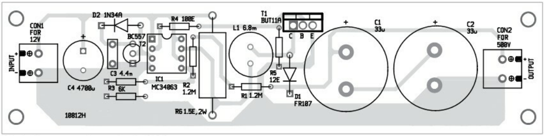

Fig. 8: Components layout for the PCB

in Run As: field. Then, click 'OK' to proceed.

HVBoostCalculator.html is HTML script and its associated image is HVcircuit.jpg. HVDesign.js is Javascript script. Run the HVBoost Calculator.html to get the page as shown in Fig. 6.

First, you need to feed 9-12V DC input and voltage tolerance depending on the supply you are using; generally, voltage tolerance is 1 per cent. Then specify the required output voltage and current in the respective form fields. (For higher output voltages, please use transistor T1 with higher voltage and current specifications.)

Using the datasheet of power transistor T1, find its Vce saturation value and put in the form field. Also get forward voltage drop of diode D1 from its datasheet feed in the form field. These parameters are very crucial for calculation of component values. After all values have been filled up in the respective fields, click 'Find Component Values' button. The form is verified for blank fields and calculation is done for the components. You will get values of R1 though R3, R6, L1, C1 and C2 as well as circuit parameters like duty cycle, switching frequency and output power.

As shown in the program output screenshot, design the circuit for 12V input voltage DC, 500V DC output, 2mA output current and 4.4nF timing capacitor. From the program output, you get values of output capacitor as 8.20µF, sensing resistor R6 as 1.59 ohms (the nearest value of 1.50 ohms) and inductor L1 as 6.8mH. The complete circuit diagram of this design is shown in Fig. 3.

Construction and testing

An actual-size PCB layout of the HVDC power supply using MC34063 is shown in Fig. 7 and its components layout in Fig. 8. Use suitable heat-sink for transistor T1. Keep inductor L1 and transistor T1 away from the main circuit section. Preferably use a shielded-type inductor for L1.

R6 should be 2W, flameproof resistor. Use the nearest value given by the software. Use proper heat-sink for power transistor T1. Keep inductor L1, transistor T1 and MC34063 away from each other on the board to minimise EMI interference.

For exact value of R3, use a parallel combination of resistors. For example, use a 6.8-kilo-ohm resistor in parallel with a 56-kilo-ohm resistor to get 6 kilo-ohms. Avoid using a trimpot because of thermal drift. Use capacitor C3 such that the frequency is within 10kHz, to avoid switching problems and heating of transistor T1.

Caution. Handle the high-voltage DC circuit with utmost care as it may cause electric shock. EFY

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at *source.efymag.com*

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